

Useful Information

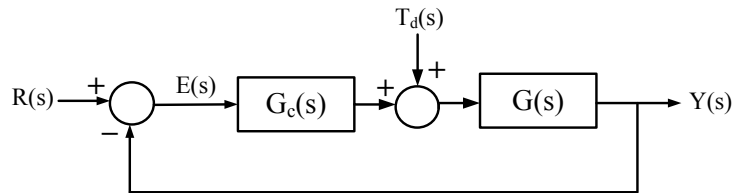
- The solutions to the quadratic equation $ax^2 + bx + c = 0$ are $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
- In the function $f(t) = Ae^{-t/\tau}$ the term τ is sometimes referred to as the time constant.
- The following Laplace transform pairs may be of use. Here $u(t)$ is the unit step function

$$\begin{aligned}
 u(t) &\longleftrightarrow \frac{1}{s} & t^n u(t) &\longleftrightarrow \frac{n!}{s^{n+1}} \\
 e^{-at} u(t) &\longleftrightarrow \frac{1}{s+a} & \frac{df(t)}{dt} &\longleftrightarrow sF(s) - f(0^-) \\
 \sin(\omega t) u(t) &\longleftrightarrow \frac{\omega}{s^2 + \omega^2} & \frac{d^2 f(t)}{dt^2} &\longleftrightarrow s^2 F(s) - sf(0^-) - f'(0^-) \\
 \cos(\omega t) u(t) &\longleftrightarrow \frac{s}{s^2 + \omega^2} & \int_{0^-}^t f(\lambda) d\lambda &\longleftrightarrow \frac{F(s)}{s} \\
 e^{-at} \cos(\omega t) u(t) &\longleftrightarrow \frac{s+a}{(s+a)^2 + \omega^2} \\
 e^{-at} \sin(\omega t) u(t) &\longleftrightarrow \frac{\omega}{(s+a)^2 + \omega^2}
 \end{aligned}$$

- A generic form for a rational Laplace Transform is $G(s) = K_G \frac{\prod_{i=1}^M (s+z_i)}{\prod_{j=1}^n (s+p_j)}$, where the set of finite zeros is $\{-z_i\}_{i=1}^M$ and the set of poles is $\{-p_j\}_{j=1}^n$.
- The final value theorem states that for a Laplace Transform $Y(s)$ with all its poles in the left half plane, except for at most one pole at the origin, $\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$.
- For a standard second-order system with a transfer function of the form $T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$, when $0 < \zeta < 1$, the poles are at positions $-\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$. The step response of such a system has the following properties:

- $y_{\text{step}}(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n \sqrt{1-\zeta^2} t + \theta)$, where $\theta = \cos^{-1}(\zeta)$.
- 2% settling time, $T_s \approx \frac{4}{\zeta\omega_n}$.
- Percentage (maximum) overshoot calculated from step response, $P.O. = 100 \left[\frac{M_{pt} - fv}{fv} \right]$, where M_{pt} is the peak value and fv is the steady-state value of the step response.
- Percentage (maximum) overshoot (given ζ), $P.O. = 100e^{-\pi\zeta/\sqrt{1-\zeta^2}}$. Hence, $\zeta = \frac{-\ln(P.O./100)}{\sqrt{\pi^2 + \ln^2(P.O./100)}}$.
- Peak time, $T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$.

- Consider a feedback control system of the following form:

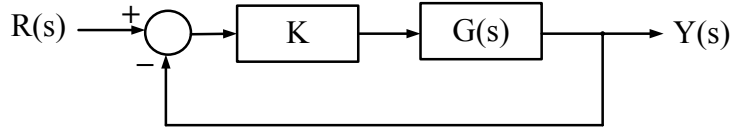


The Laplace Transform of the error is $E(s) = \frac{1}{1+G_c(s)G(s)} R(s) - \frac{G(s)}{1+G_c(s)G(s)} T_d(s)$

$$dB = 20 \log_{10} X \quad , \quad X = \text{linear gain}$$

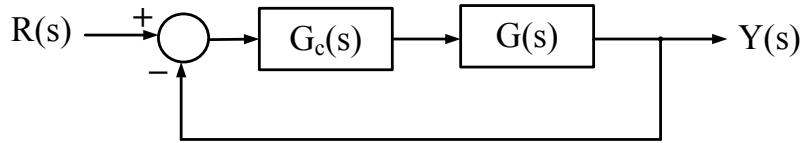
$$X = 10^{\frac{dB}{20}}$$

- Steady state error constants:
 - Position error constant: $K_{\text{posn}} = \lim_{s \rightarrow 0} G_c(s)G(s)$
 - Velocity error constant: $K_v = \lim_{s \rightarrow 0} sG_c(s)G(s)$
- Consider a feedback control system of the following form:



where $K > 0$ and $G(s)$ takes the form $G(s) = \frac{K_G \prod_{i=1}^M (s+z_i)}{\prod_{j=1}^n (s+p_j)}$, with $K_G > 0$.

- The point s_0 is on the root locus if $\sum_{i=1}^M \angle(s_0 + z_i) - \sum_{j=1}^n \angle(s_0 + p_j) = 180^\circ + k360^\circ$ for some integer k .
- The value of the gain K such that s_0 is a pole of the closed loop is $K = \frac{\prod_{j=1}^n |s_0 + p_j|}{K_G \prod_{i=1}^M |s_0 + z_i|}$.
- Consider the following closed-loop system (where the open-loop transfer function is $L(s) = G_c(s)G(s)$):



- The gain margin is $K_{\text{gm}} = \frac{1}{|L(j\omega_x)|} = \frac{1}{|G_c(j\omega_x)G(j\omega_x)|}$, where ω_x is the frequency at which $\angle G_c(j\omega_x)G(j\omega_x) = -180^\circ$.
- K_{gm} (in dBs) = $-20 \log_{10}(|L(j\omega_x)|)$
- The phase margin is $\phi_{\text{pm}} = 180^\circ + \angle L(j\omega_c) = 180^\circ + \angle G_c(j\omega_c)G(j\omega_c)$, where the crossover frequency ω_c is the frequency at which $|G_c(j\omega_c)G(j\omega_c)| = 1$.
- In the frequency domain, for a closed-loop system of the form $T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$, the 3dB bandwidth is:
 - $\omega_{\text{cl,3dB}} = \omega_n (\sqrt{2 - 4\zeta^2 + 4\zeta^4} + 1 - 2\zeta^2)^{1/2}$
- In the frequency domain, for an open-loop system of the form $L(s) = G_c(s)G(s) = \frac{\omega_n^2}{s(s+2\zeta\omega_n)}$:
 - The phase margin is: $\phi_{\text{pm}} = \tan^{-1}\left(\frac{2}{\sqrt{(4+1/\zeta^4)^{1/2}-2}}\right)$. If ϕ_{pm} is measured in degrees, and if $\zeta \leq \frac{1}{\sqrt{2}}$, then $\phi_{\text{pm}} \approx 100\zeta$.
 - The crossover frequency is: $\omega_c = \omega_n (\sqrt{1 + 4\zeta^4} - 2\zeta^2)^{1/2}$.
 - The 2% settling time is: $T_s = \frac{4f_c(\zeta)}{\omega_c}$, where $f_c(\zeta) = (\sqrt{4 + 1/\zeta^4} - 2)^{1/2}$.
- A first order lead compensator has the form $G_c(s) = \frac{K_c(s+z)}{s+p}$ with $|z| < |p|$ and $\alpha_{\text{lead}} = \frac{p}{z}$. The peak of the phase response of $G_c(j\omega)$ occurs at $\omega = \omega_m = \sqrt{zp}$. That peak, $\phi_m = \angle G_c(j\omega_m)$, satisfies $\sin(\phi_m) = \frac{\alpha_{\text{lead}} - 1}{\alpha_{\text{lead}} + 1}$, which means $\alpha_{\text{lead}} = \frac{1 + \sin(\phi_m)}{1 - \sin(\phi_m)}$.
- A first order lag compensator has the form $G_c(s) = \frac{K_c(s+z)}{s+p}$ with $|p| < |z|$ and $\alpha_{\text{lag}} = \frac{z}{p}$.